Appendix 2. Meta-analyses by diagnostic grouping

Results

To assist with the interpretation of the overall results, additional analyses were performed to examine each diagnostic group with at least three included studies whose data were able to be meta-analysed (Spinal cord injury, Multiple Sclerosis, Duchenne Muscular Dystrophy, Parkinson' disease, and Amyotrophic lateral sclerosis). The FVC, MIP and MEP data were chosen to illustrate the most commonly observable effects and to allow consideration of heterogeneity across all studies versus heterogeneity attributable to diagnoses *per se*. These data are summarized in the Table below and illustrated in the series of Forest Plots

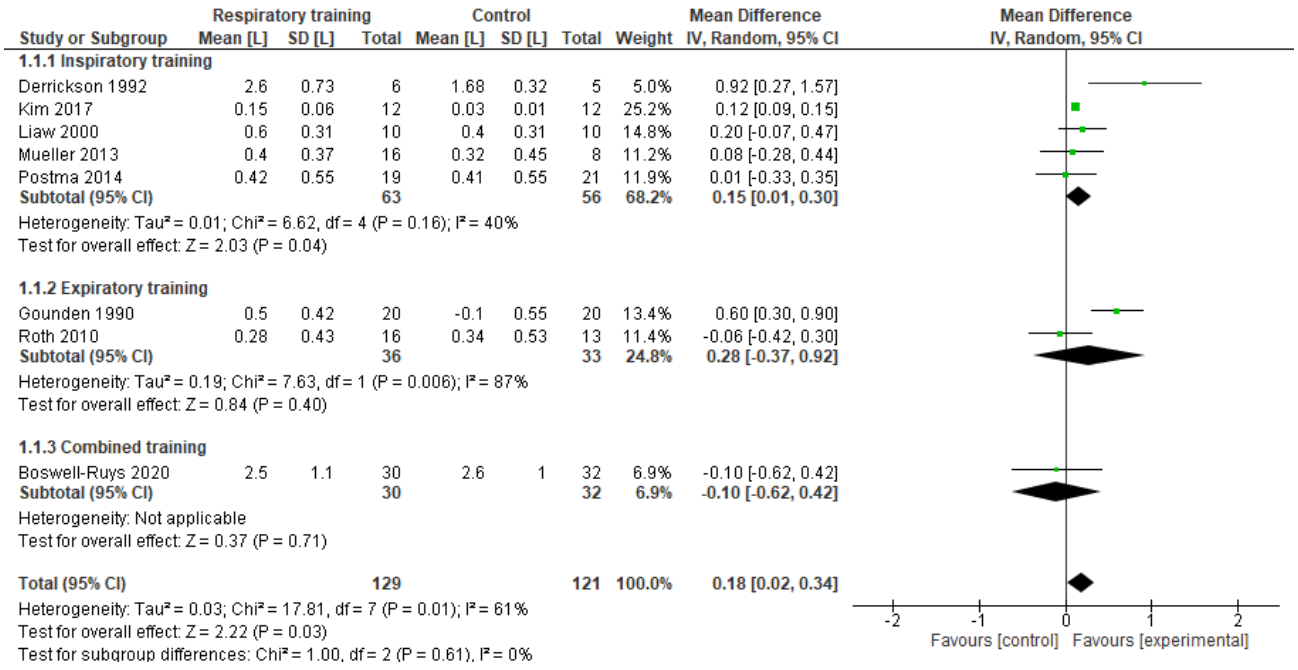
Online Table 4: Heterogeneity and standardised mean differences for overall effect estimates of forced vital capacity, maximal expiratory, and inspiratory pressures across diagnoses.

Diagnoses & total number of included studies	Number of studies per outcome in effect estimates	Outcome variable	I^2 (p-value)	Standardised mean difference
Spinal Cord Injury	8	FVC	**61%, 0.01**	**0.18**
15 studies	8	MIP	**58%, 0.02**	**9.48**
	7	MEP	**74%, <0.01**	**9.18**
Multiple sclerosis	2	FVC	0%, 0.53	0.17
5 studies	4	MIP	0%, 0.76	2.11
	4	MEP	**70%, 0.02**	5.84
Duchenne muscular dystrophy	2	FVC	0%, 0.78	0.08
4 studies	1	MIP	NA	1.3
	0	MEP		
Parkinson's disease	2	FVC	0%, 0.58	0.33
3 studies	2	MIP	0%, 0.86	10.65
	3	MEP	0%, 0.55	**31.33**
Amyotrophic lateral sclerosis	1	FVC	NA	0.70
3 studies	1	MIP	NA	6.8
	1	MEP	NA	28.9

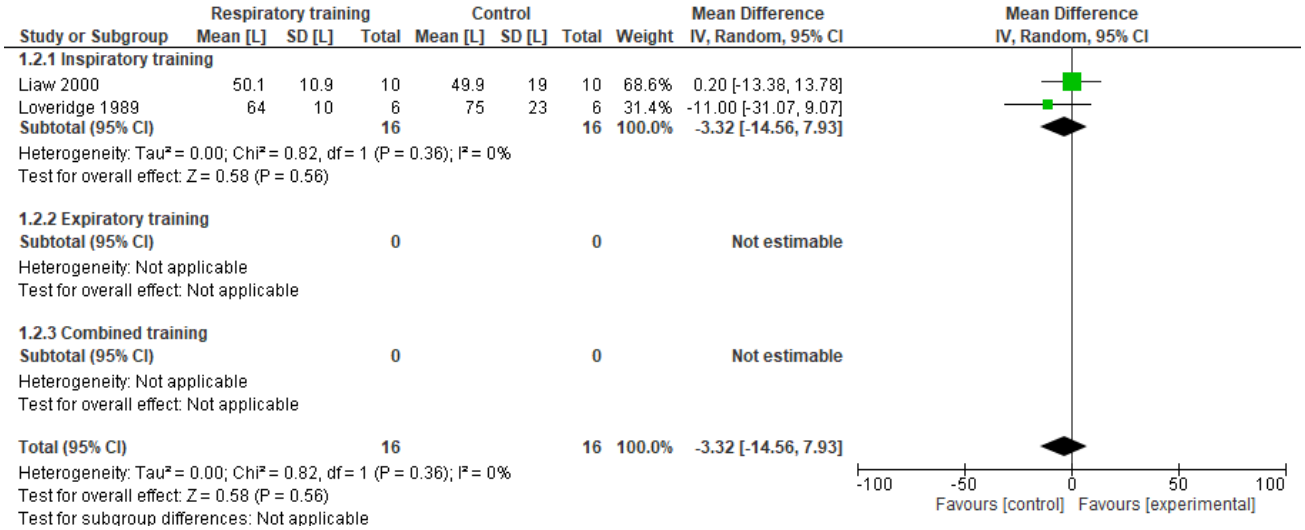
Bold indicates that the standardised mean difference favoured the intervention and the confidence intervals did not cross zero. FVC = forced vital capacity, %pred = measure as a percentage of predicted, NE = Not estimable, MIP = maximal inspiratory pressure, MEP = maximal expiratory pressure, SNIP = sustained nasal inspiratory pressure, PEF = peak expiratory flow, PCF = peak cough flow.

Online Figure 4: Forest plots for all available meta-analysable data in Spinal cord injury, Multiple Sclerosis, Duchenne Muscular Dystrophy, Parkinson' disease, and Amyotrophic lateral sclerosis

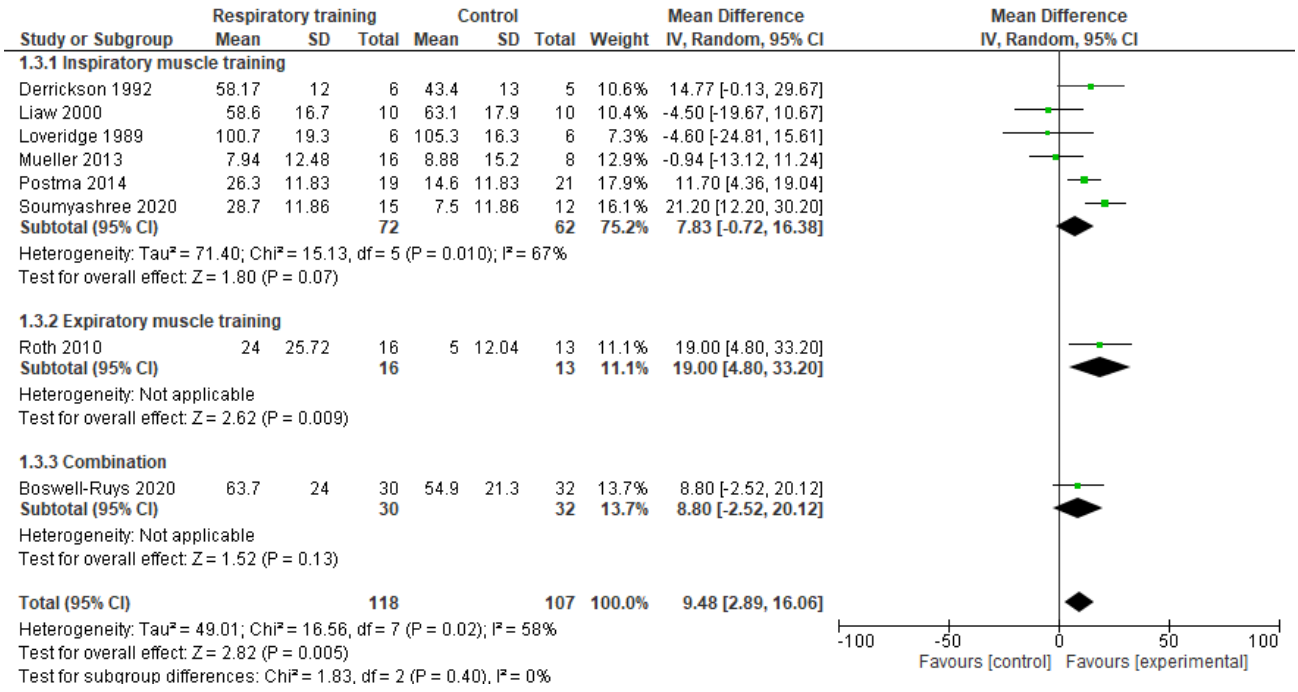
Spinal Cord Injury



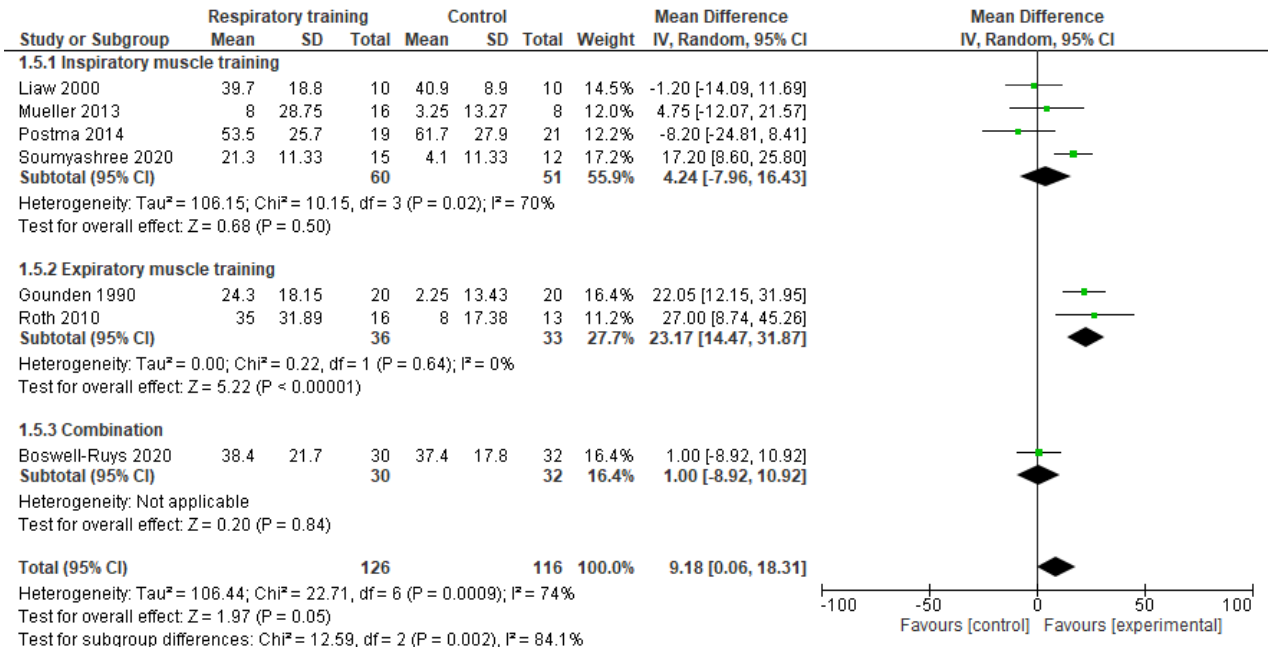
Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.1 Forced Vital Capacity (L) [L].



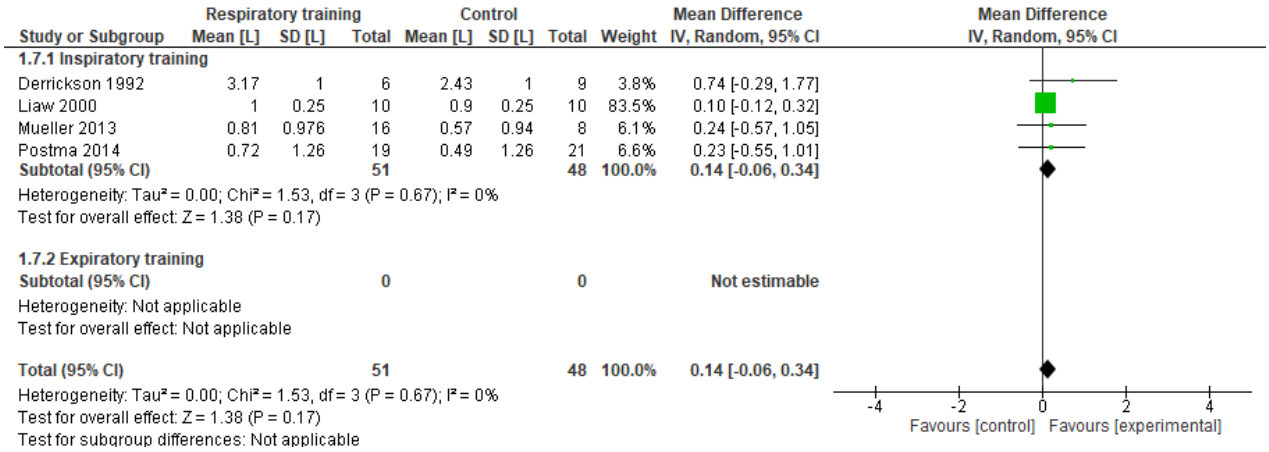
Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.2 Forced Vital Capacity (% predicted) [L].



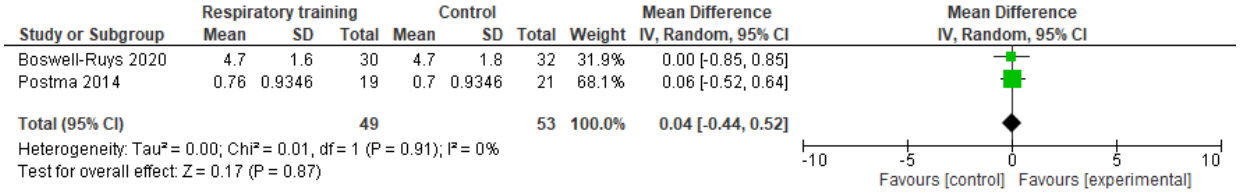
Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.3 Maximum inspiratory pressure.



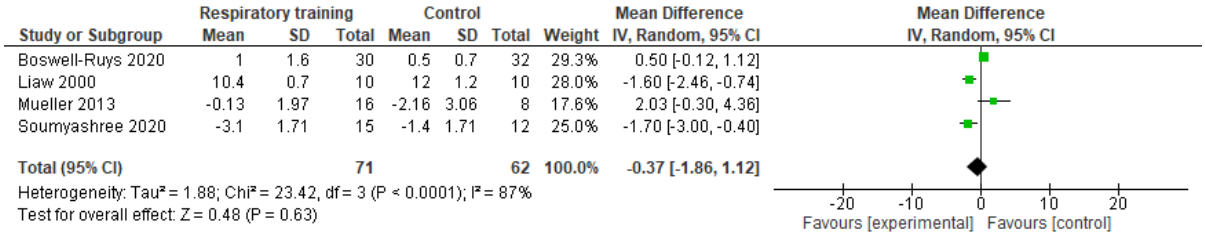
Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.5 Maximum expiratory pressure.



Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.7 Peak expiratory flow [L].



Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.9 Peak cough flow.

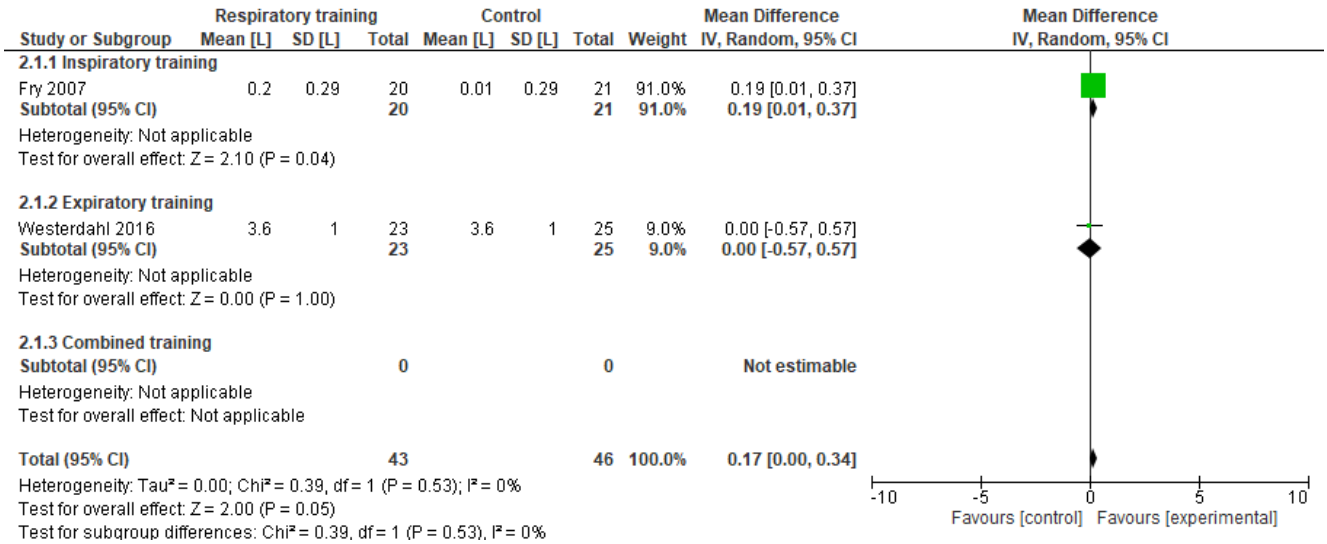


Forest plot of comparison: 1 SCI - Respiratory muscle training vs Control, outcome: 1.10 Dyspnoea.

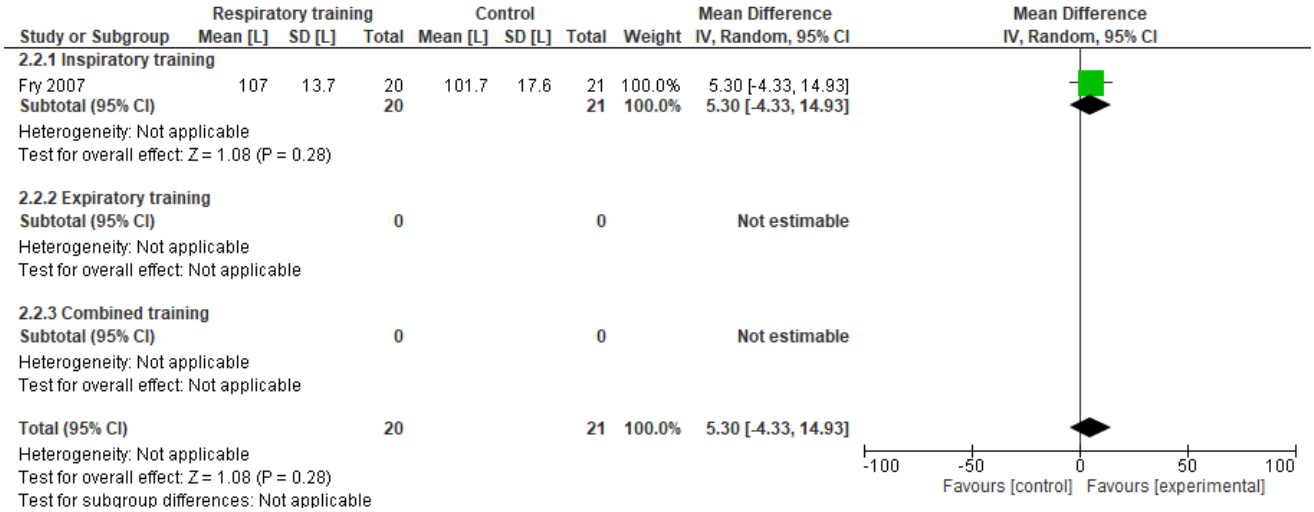
No data:

- MIPs % Pred
- MEP % Pred

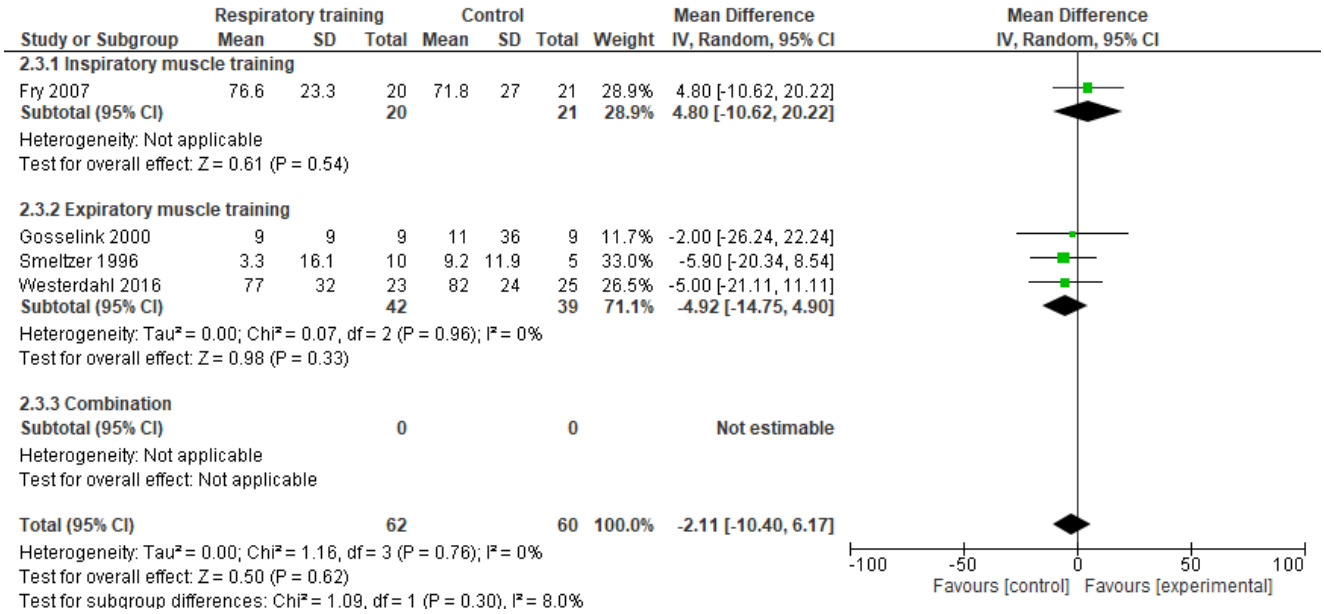
Multiple Sclerosis



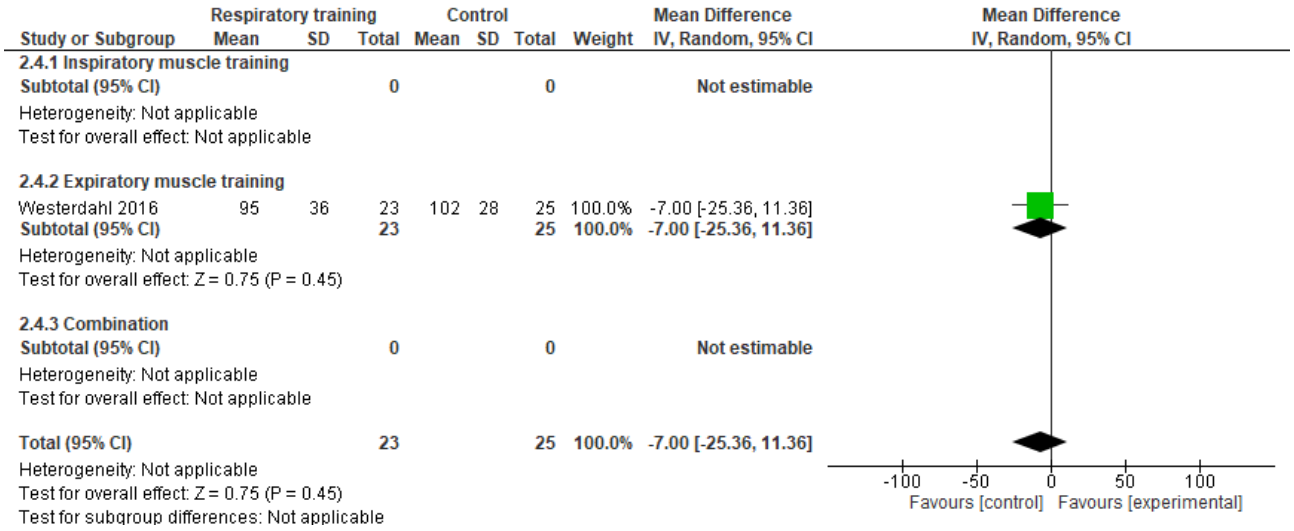
Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.1 Forced Vital Capacity (L) [L].



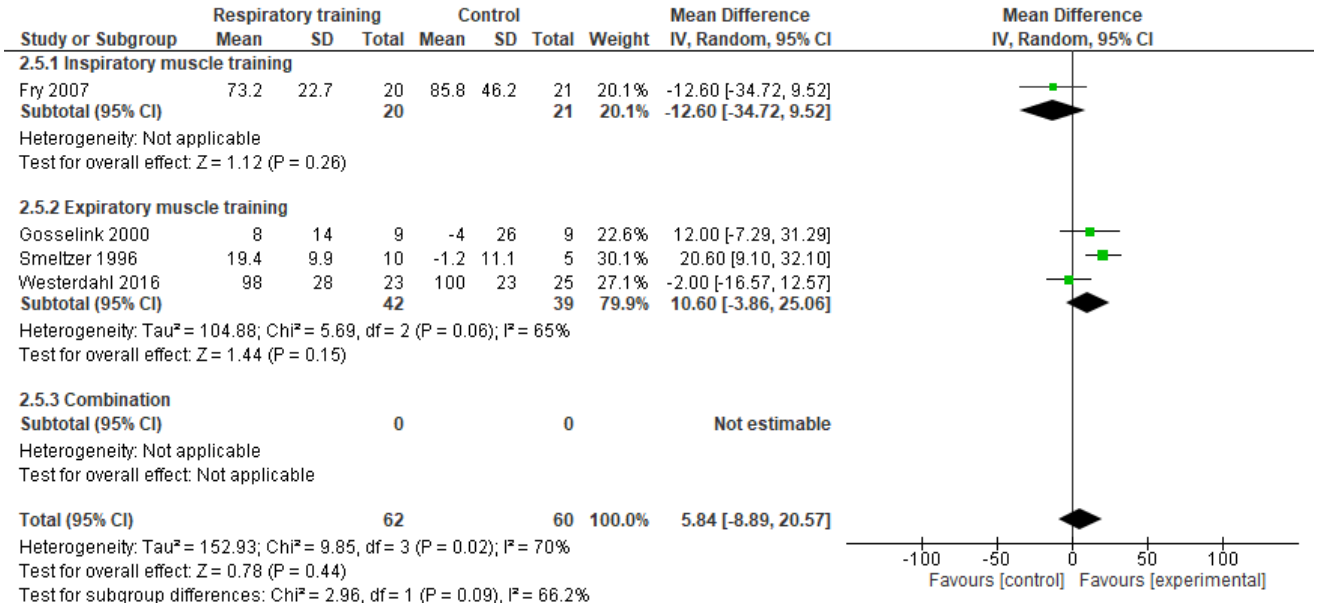
Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.2 Forced Vital Capacity (% predicted) [L].



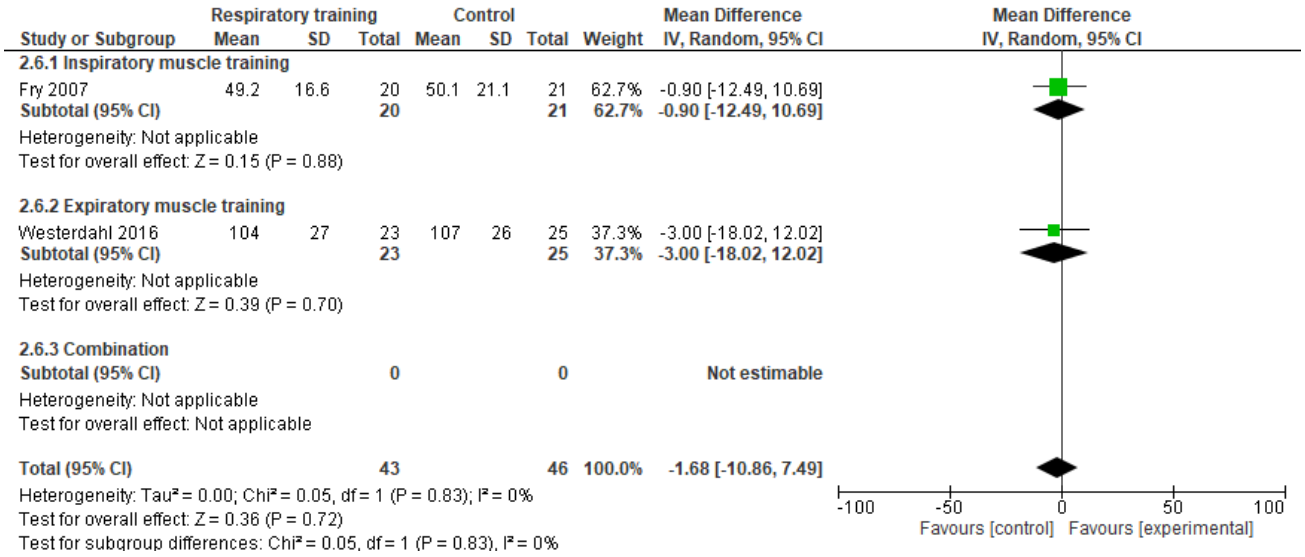
Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.3 Maximum inspiratory pressure.



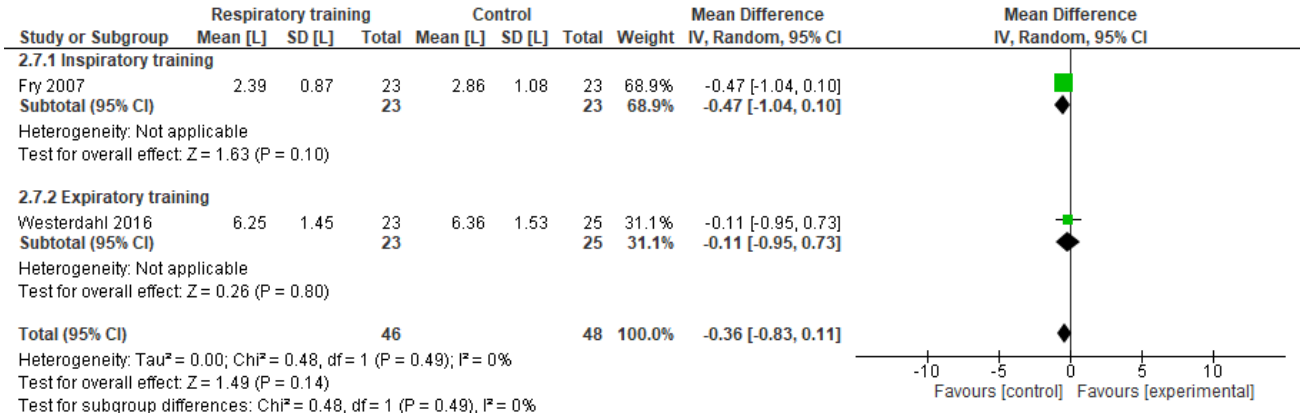
Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.4 Maximum inspiratory pressure (% predicted).



Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.5 Maximum expiratory pressure.



Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.6 Maximum expiratory pressure (% predicted).

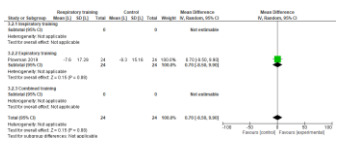


Forest plot of comparison: 2 MS - Respiratory muscle training vs control, outcome: 2.7 Peak expiratory flow [L].

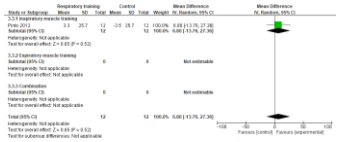
No Data:

- PCF
- Dyspnoea

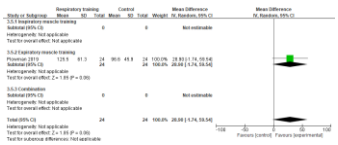
Motor Neurone Disease / Amyotrophic Lateral Sclerosis



Forest plot of comparison: 3 ALS/MND - Respiratory muscle training vs Control, outcome: 3.2 Forced Vital Capacity (% predicted) [L].



Forest plot of comparison: 3 ALS/MND - Respiratory muscle training vs Control, outcome: 3.3 Maximum inspiratory pressure.



Forest plot of comparison: 3 ALS/MND - Respiratory muscle training vs Control, outcome: 3.5 Maximum expiratory pressure.

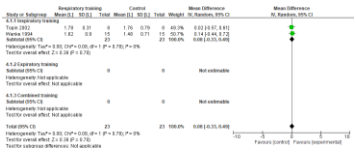


Forest plot of comparison: 3 ALS/MND - Respiratory muscle training vs Control, outcome: 3.9 Peak cough flow.

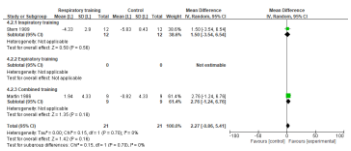
No data:

- FVC (L)
- MIP (% pred)
- MEP (% pred)
- PEF
- Dyspnoea

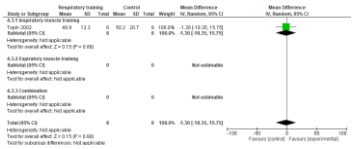
Duchenne Muscular Dystrophy



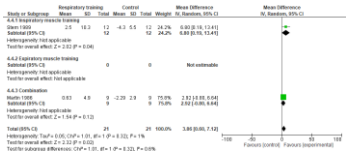
Forest plot of comparison: 4 DMD - Respiratory muscle training vs Control, outcome: 4.1 Forced Vital Capacity (L) [L].



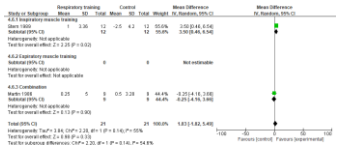
Forest plot of comparison: 4 DMD - Respiratory muscle training vs Control, outcome: 4.2 Forced Vital Capacity (% predicted) [L].



Forest plot of comparison: 4 DMD - Respiratory muscle training vs Control, outcome: 4.3 Maximum inspiratory pressure.



Forest plot of comparison: 4 DMD - Respiratory muscle training vs Control, outcome: 4.4 Maximum inspiratory pressure (% predicted).

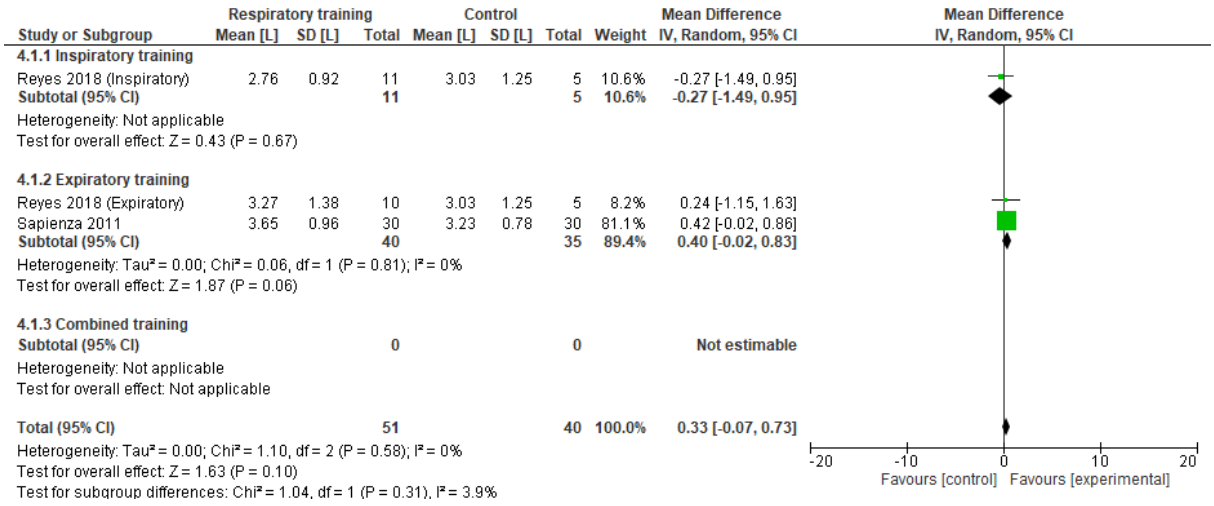


Forest plot of comparison: 4 DMD - Respiratory muscle training vs Control, outcome: 4.6 Maximum expiratory pressure (% predicted).

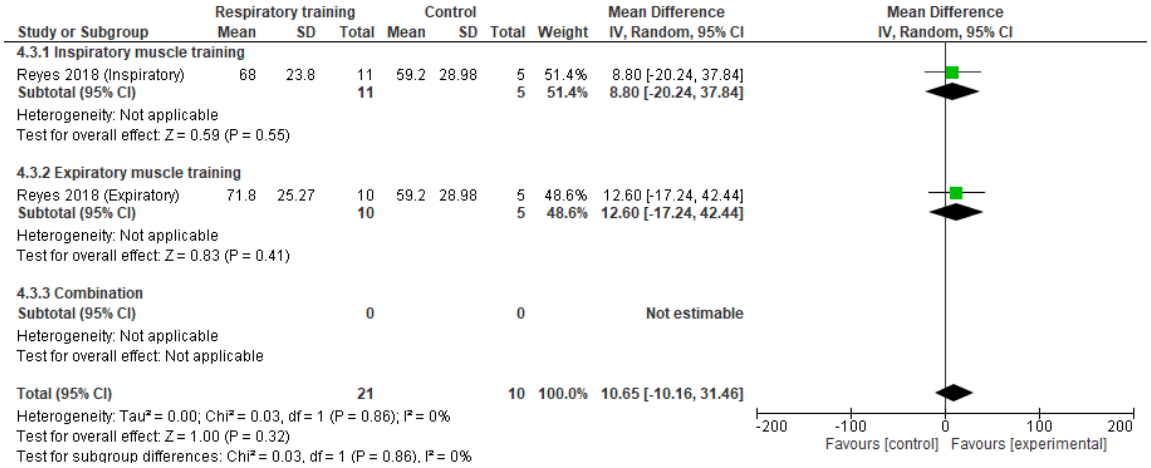
No Data:

- MEPs
- Peak expiratory flow
- PCF
- Dyspnoea

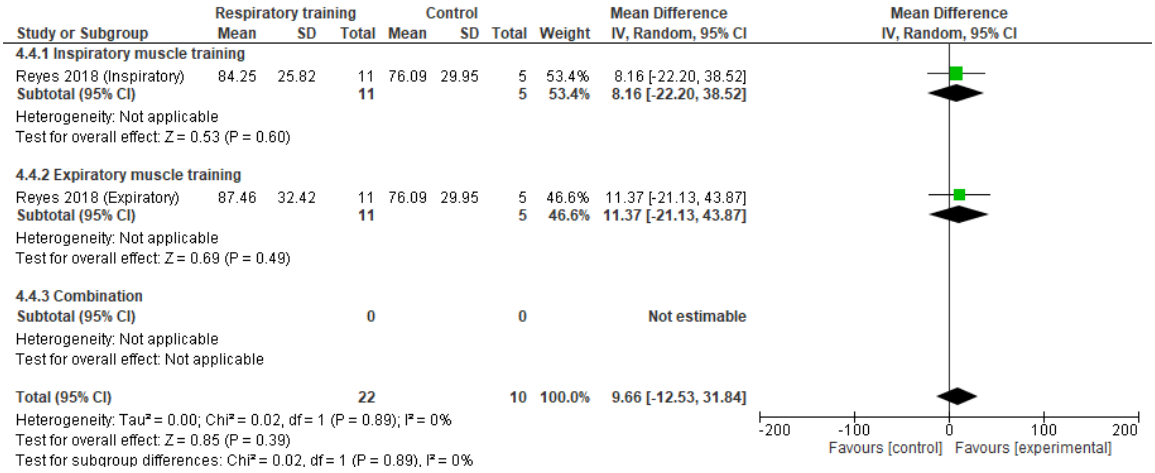
Parkinson's Disease



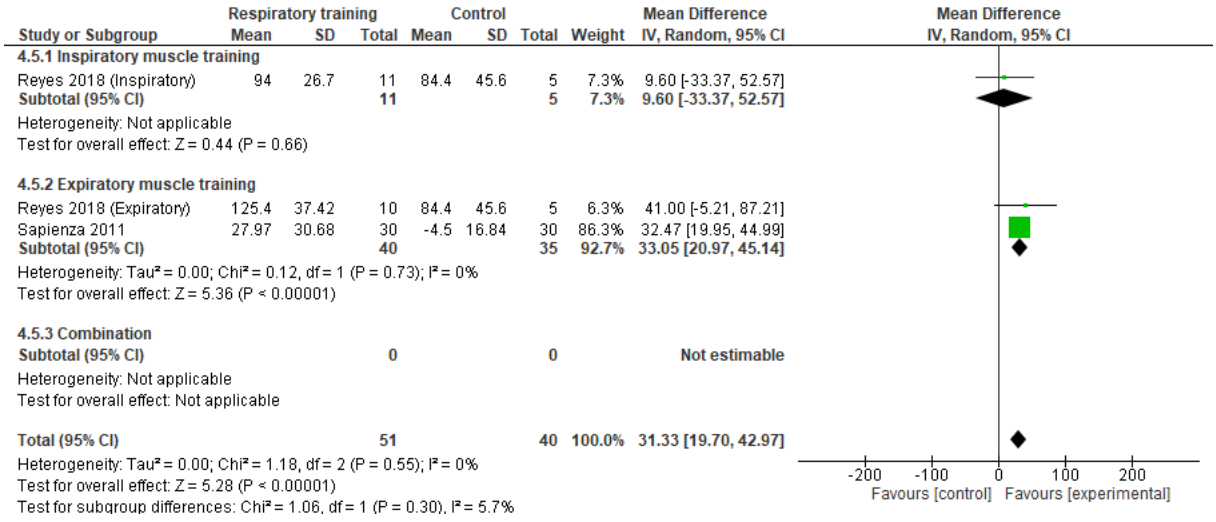
Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.1 Forced Vital Capacity (L) [L].



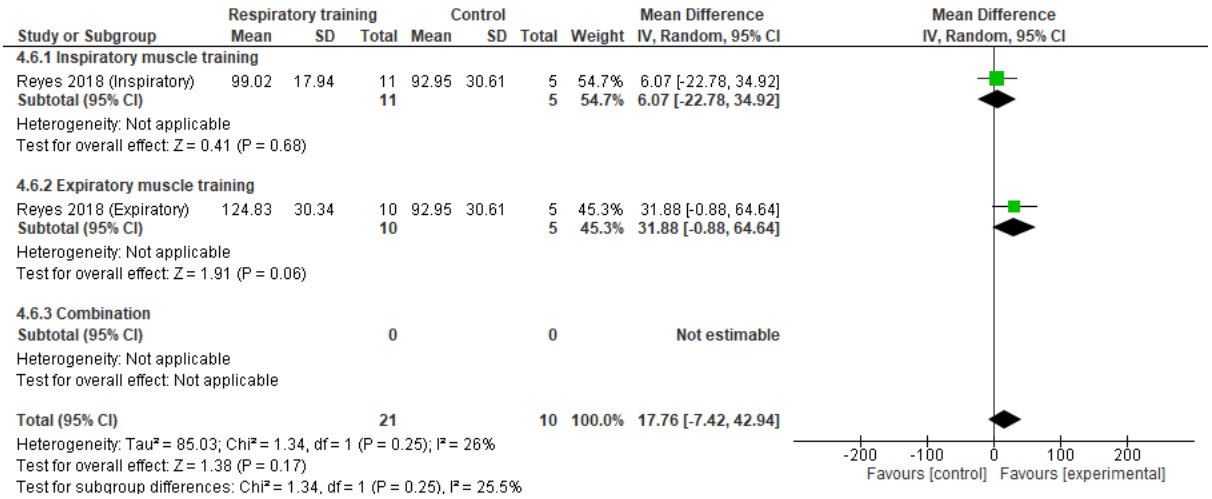
Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.3 Maximum inspiratory pressure



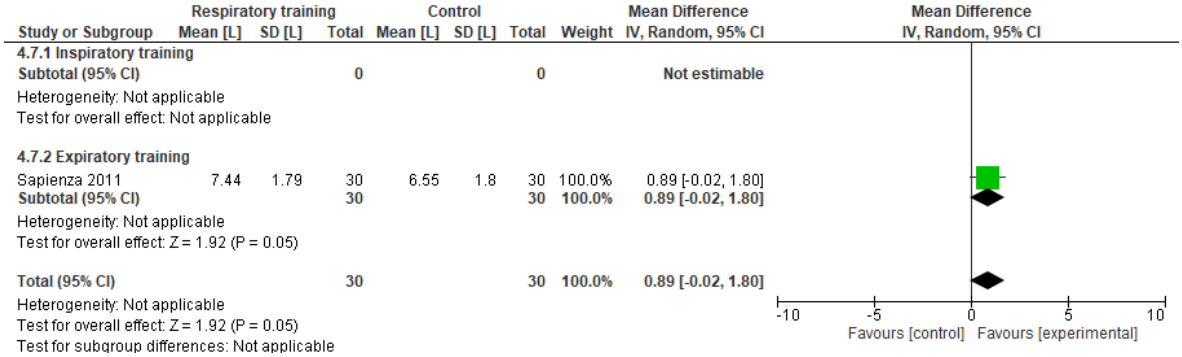
Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.4 Maximum inspiratory pressure (% predicted).



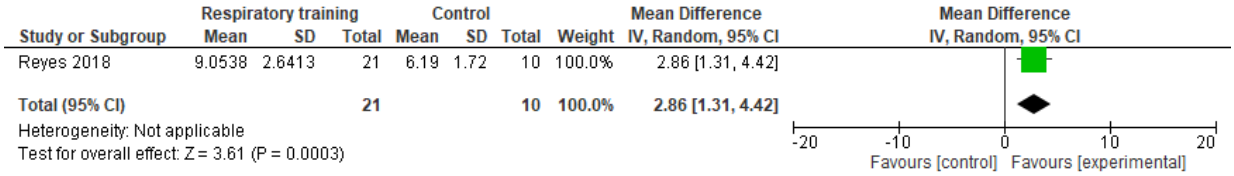
Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.5 Maximum expiratory pressure.



Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.6 Maximum expiratory pressure (% predicted).



Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.7 Peak expiratory flow [L].



Forest plot of comparison: 4 PD - Respiratory Muscle Training vs Control, outcome: 4.9 Peak cough flow.

No data:

- FVC % pred